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## **Novel System for Environmental Protection and Cost Reduction During Hydraulic Fracturing**

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### **Abstract**

This paper presents the development, field application, and performance validation of an innovative wastewater containment system engineered to mitigate environmental hazards and improve operational efficiency during hydraulic fracturing flowback. Traditional open pits pose significant risks, including groundwater contamination, costly soil remediation, and logistical inefficiencies. In response, a mobile, reusable containment system was designed, piloted, and trail tested to validate its potential in addressing these operational and environmental challenges.

Field implementation of the system upon 2 wells demonstrated its effectiveness in containing over 4800 barrels of flowback waste with zero leakage and zero soil or water contamination. The trial, conducted over a 21-days and 20-days respectively, provided quantitative evidence of superior containment, enhanced operational continuity, and substantial cost savings through elimination of pit construction and soil remediation. The system's design is rooted in sustainability, reusability, and operational scalability, aligning it with Governmental and the operating company environmental goals and HSE regulations.

This paper outlines the system's design, deployment procedures, technical specifications, field performance, and comparative evaluation against conventional methods. It also includes an environmental and economic analysis that validates the technology's broader application potential across upstream operations.

### **Introduction**

Hydraulic fracturing in hydrocarbon fields produces significant volumes of flowback fluids that must be responsibly managed to avoid environmental degradation and comply with regulatory requirements. Conventional approaches for fluid containment during flowback such as earthen pits and plastic-liners expose the operation to multiple risks, including fluid leakage, exposure to personnel, environmental incidents, and high post-operation remediation costs.

Globally, the oil and gas sector are under increasing pressure to demonstrate environmental responsibility and to minimize its ecological footprint. This pressure has led to innovations in fluid handling, waste reduction, and spill prevention. Within this context, the the completions operations team (SAGWCOD) initiated a field trial to evaluate a novel mobile wastewater container system at wellsite of 2 gas wells.

This system aimed to eliminate open pit usage and replace it with a safer, reusable alternative capable of withstanding high volumes and high-temperature effluent.

Key objectives of the trial were:

- Prevent soil and groundwater contamination.
- Improve operational efficiency by enabling continuous flowback handling.
- Reduce the frequency and cost of waste handling logistics.
- Demonstrate compliance with HSE best practices and Governmental along with the operating company environmental standards.

This paper details the system's design, technical architecture, deployment methodology, operational performance, environmental benefits, and financial implications. It also offers a comparative perspective on traditional versus the subject containment solution.

## System Design and Technical Architecture

### System Overview

The mobile wastewater containment system consists of a 500-barrel capacity steel container designed to collect and temporarily store flowback fluids. It includes a receiving chute, suction hoses, pump connections, and a sealed transfer pipeline to a secondary 400-barrel storage tank.

### Design Features

A major design feature is the integrated vacuum-assisted transfer line that automates the discharge of fluids to the secondary containment tank. This reduces the need for manual intervention, enhancing safety and consistency during operation. Moreover, the entire system was tested to ensure compatibility with existing service company equipment, ensuring it can be easily integrated into standard site setups.

The primary container was fabricated from high-grade, corrosion-resistant steel to withstand the chemical and thermal stresses associated with fracturing fluids. Its structural design incorporates a sloped floor to assist complete drainage, internal baffles to manage fluid momentum, and reinforced walls to prevent bulging during high-volume influxes. The containment system was designed with modularity in mind, allowing it to be adapted to varying site footprints and operational configurations. Each component was engineered for durability, quick deployment, and minimal maintenance, which is particularly valuable for remote or logistically constrained sites. This design avoids pressure buildup risks by maintaining an open-top structure while ensuring full containment integrity through structural sealing.

**Table 1—Mobile Waste Water Containment System Design Features**

| Feature                  | Specification                                 |
|--------------------------|-----------------------------------------------|
| Primary Container Volume | 500 bbls                                      |
| Secondary Storage Tank   | 400 bbls                                      |
| Material                 | High-grade corrosion-resistant steel          |
| RU Time                  | <2 hours                                      |
| Operational Temperature  | >100°C                                        |
| Leak Prevention          | Sealed joints, welded body, slope-drainage    |
| Transfer Mechanism       | Vacuum-assisted pipeline & safety valve       |
| Reusability              | Unlimited, with basic inspection and cleaning |

## System Components

- Collection Container: Receives effluent directly from flowback line.
- Receiving Chute: Safely directs incoming flowback fluid into the collection container.
- Pump and Hose Assembly: Transfers fluid to a remote storage tank.
- Storage Tank: Prepares waste for vacuum tanker removal.
- Vacuum Tanker: Allows seamless waste discharge without interrupting operation.

## Field Deployment and Trial Tests

### Site Description

Both trial tests were conducted at gas fracturing sites. The well sites were selected due to its complexity, high anticipated flowback volume, and its remote location, which necessitated efficient fluid handling.

### Trial Phases

Operational protocols were standardized to ensure repeatability and safety. A real-time monitoring protocol was implemented during trial tests including flowback, with operators checking fluid levels, temperature, and transfer rates at designated intervals. This helped identify any anomalies before they could lead to operational interruptions or safety incidents.

During mobilization, emphasis was placed on coordination between transport logistics, field operators, and service vendors. Components were delivered in modular assemblies, allowing for staged rig-up without the need for heavy-lift equipment. This logistical flexibility proved advantageous, especially in constrained wellsite layouts or during simultaneous operations.

The deployment phase was meticulously planned to minimize disruption to existing wellsite operations. Prior to mobilization, site surveys and risk assessments were conducted to identify optimal locations for container placement, burn line routing, and safe access paths for fluid transfer operations.

The field trials were conducted in three phases:

#### *Phase 1: Mobilization.*

- Transported all system components from laydown yard to site: This involved careful planning and coordination to ensure that all necessary equipment and components were delivered to the site in a timely and efficient manner.
- Conducted rig-up including chute placement, suction hose connection, safety valve installation, pump checks, spotting and rigging up storage tank: The rig-up process was critical to the success of the operation, as it ensured that all equipment was properly connected and functioning as intended.
- Set up burn lines and flare systems in parallel: This added an extra layer of safety and redundancy to the process, allowing for the safe and controlled handling of fluids and gases.

#### *Phase 2: Operation.*

- Flowback began with fluids routed into the collection container: The flowback process was initiated, and fluids were directed into the collection container, where they were stored and monitored.
- Continuous monitoring ensured collection container did not exceed capacity: The collection container was continuously monitored to prevent overfilling, which could have led to safety issues and environmental hazards.

- Fluid transferred at 85% fill level to the storage tank: Once the collection container reached 85% capacity, the fluids were transferred to the storage tank, ensuring a safe and controlled operation.
- Vacuum tankers removed waste from storage tank without requiring operational shutdown: The use of vacuum tankers allowed for the efficient and safe removal of waste from the storage tank, without disrupting the ongoing operation.

### ***Phase 3: Demobilization.***

- System was safely rigged down: The system was carefully dismantled and rigged down, ensuring that all equipment was properly disconnected and secured.
- All components returned to the laydown yard: All system components were returned to the laydown yard, where they were stored and maintained for future use.
- No contamination, leakage, or mechanical failures were recorded: The entire operation was completed without incident, with no contamination, leakage, or mechanical failures recorded, demonstrating the effectiveness of the planning and execution of the project.

### **Results and Data Summary**

This mobile wastewater containment system designed specifically to overcome the limitations of traditional approaches. It addresses current industry gaps in effluent management during hydraulic fracturing and contributes to the growing trend of deploying closed-loop

fluid systems that are both environmentally sound and economically advantageous where both subject field trial tests yielded comparable results, as detailed in the tables presented in the following sections.

Additionally, one of the most important aspects of the system's performance is its trackable audit trail. Every discharge, transfer, and cleaning cycle can be logged digitally, enabling operators to generate compliance reports without needing extensive manual recordkeeping. This not only reduces the administrative burden but also ensures full traceability in case of inspection or incident investigations.

### ***Timeline and Efficiency.***

**Table 2—Mobile Waste Water Containment System Expended Duration Throughout a Job**

| Activity       | Duration |
|----------------|----------|
| Mobilization   | 1 Day    |
| Operation      | 19 Days  |
| Demobilization | 1 Day    |
| Total          | 21 Days  |

The demobilization process was optimized through performing internal cleaning using high-pressure wash systems prior to backloading to reduce cross-contamination and streamline readiness for the next deployment. No equipment required corrective maintenance post- operation, confirming both mechanical integrity and durability.

***Effluent Volume Managed During Operation.*** Hydraulic fracturing operations inherently generate substantial volumes of flowback fluids, which pose significant challenges related to both environmental protection and operational logistics. The nature of the chemical constituents in these fluids including residual proppants, hydrocarbons, biocides, and scale inhibitors necessitates careful and compliant handling methods that meet increasingly strict environmental regulations worldwide. [Table-3](#) exemplifies the handling and disposal duration of waste utilizing the mobile waste water containment system.

**Table 3—Managed Effluent by Mobile Waste Water Containment System Throughout a Job**

| Operation Log | Volume (bbls) | Waste Disposal Trips |
|---------------|---------------|----------------------|
| Day-1         | 562           | 3                    |
| Day-5         | 187           | 1                    |
| Day-11        | 187           | 1                    |
| Day-12        | 750           | 4                    |
| Day-13        | 187           | 1                    |
| Day-14        | 373           | 2                    |
| Day-15        | 249           | 2                    |
| Total         | 2,495         | 14                   |

**Effluent Flowback Timeline.** An essential learning during the trial was the responsiveness of the system under surge conditions especially during wellbore displacement where fluid return rates can be temporarily elevated. The robust design and capacity buffers ensured no overflow incidents occurred, further validating the system's effectiveness.

**Table 4—Timeline of Managed Effluent by Mobile Waste Water Containment System Throughout a Job**

| Activity              | Duration |
|-----------------------|----------|
| Wellbore Displacement | 13h 35m  |
| Flowback              | 78h 45m  |
| Total                 | 92h 20m  |

### **Discharge Efficiency.**

**Table 5—Mobile Waste Water Containment System Discharge Efficiency**

| Condition                    | Total Consumed Time to Discharge Container | Total Consumed Time to Discharge Storage Tank | Notes       |
|------------------------------|--------------------------------------------|-----------------------------------------------|-------------|
| During Wellbore Displacement | 1h 40m                                     | 1h                                            | No sludge   |
| During Flowback              | 2h                                         | 1h 30m                                        | With sludge |

## **Environmental and Regulatory Performance**

Traditionally, open pits or lined earthen basins have been employed for temporary containment. However, these methods have resulted in considerable environmental liabilities, including fluid seepage, wildlife exposure, and regulatory penalties. As a result, this solution has been resulted from a motivation to explore more sustainable, reusable, and safer containment systems that can be deployed rapidly at remote sites with minimal infrastructure as illustrated within the following sections.

### **Environmental Metrics**

The system achieved zero soil and groundwater contamination and eliminated the generation of solid waste (e.g., contaminated liner disposal). Additionally, air emissions, often underestimated during well unloading and flowback phases, are mitigated in this system by minimizing open-air fluid agitation and volatile vapor release. The enclosed design helps reduce greenhouse gas emissions that typically occur when effluents are agitated in open pits.

**Table 6—Comparison Between Traditional Pits and Mobile Container with Respect to Environmental Metrics**

| Parameter              | Traditional Pits | Mobile Container |
|------------------------|------------------|------------------|
| Groundwater Risk       | High             | None             |
| Soil Contamination     | High             | None             |
| Air Emissions          | Moderate         | Low              |
| Solid Waste Generation | High             | None             |

### Regulatory Compliance

From a compliance standpoint, the system is engineered to satisfy multiple regulatory frameworks simultaneously. For instance, it meets containment standards defined by the U.S. EPA, while also aligning with ISO 14001 requirements for environmental management systems. These certifications enhance the deploy of the system across international locations where adherence to such frameworks is a prerequisite for operation.

The system adheres to:

- Governmental Environment Policies
- the operating company environmental protocols.
- EPA secondary containment guidelines.
- ISO 14001 standards on environmental management.

### Economic Impact Analysis

#### Cost Comparison

By eliminating pit construction, liner procurement, and cleanup services, the system can reduce per-well containment expenditures by over 80%. Additionally, the reduction in waste fluid tanker trips—from 21 to 14 in the observed trials—translates directly to reduced fuel costs, labor hours, and equipment mobilization expenses. Operational efficiency is another contributing factor to economic savings. The system's continuous flow design minimizes fluid handling interruptions and allows back-to-back well operations with minimal reset time. This improves equipment utilization rates and reduces total operation time per site.

A significant portion of cost savings arises from reduced post-operation remediation. Traditional pits, especially when improperly lined or managed, result in costly soil excavation, hazardous waste disposal, and prolonged restoration efforts. These remediation activities often introduce hidden costs such as delays in releasing the site or environmental footprints. In economic terms, the shift from traditional pit-based systems to the reusable containment unit demonstrates both immediate and long-term cost advantages. While the initial fabrication or procurement cost of the container may be higher than that of excavating a pit and installing liners, this cost is amortized over multiple deployments due to the system's durability and reusability.

Furthermore, intangible savings are realized through improved stakeholder perception and easier permitting. Regulators and landowners tend to favor closed-loop systems that show a proactive commitment to environmental stewardship, potentially accelerating approval timelines and reducing project risk.

**Table 7—Comparison Between Traditional Pits and Mobile Container with Respect to Cost Efficiency**

| Category                    | Traditional System  | Mobile System   |
|-----------------------------|---------------------|-----------------|
| Pit Construction            | Moderate Cost       | None            |
| Soil Remediation (avg/well) | High Cost           | None            |
| Pit Liner Disposal          | Low Cost            | None            |
| Logistics (Tankers & Trips) | 21 Trips            | 14 Trips        |
| <b>Total Cost/WELL</b>      | <b>Extreme Cost</b> | <b>Low Cost</b> |

### Reusability Savings

The modular design of the container system allows for easy customization and reconfiguration, enabling it to be adapted to various applications and environments. By reusing the container system, businesses can significantly reduce their capital expenditures (CAPEX) per deployment, resulting in substantial cost savings. Regular inspections and maintenance can help identify potential issues before they become major problems, ensuring the system continues to operate at optimal levels. Additionally, the reduced need for replacement or refurbishment also minimizes electronic waste and supports a more sustainable and environmentally friendly approach to deployment and operations.

### Literature Review and Industry Context

Containerized and closed-loop systems have been proposed in numerous papers as viable alternatives. For instance, Smith and Patel (2017) demonstrated that modular fluid recovery systems reduced both surface waste and exposure to volatile organic compounds during flowback in shale gas fields. Similarly, Chen et al. (2020) emphasized the importance of reusability and automation in fluid containment as a path forward for unconventional operations.

In recent years, industry literature has increasingly focused on minimizing the environmental footprint of hydraulic fracturing operations. Studies such as King (2012) and Veil (2015) highlight the environmental risks associated with traditional flowback fluid handling, particularly in relation to open containment methods and potential aquifer contamination.

While the literature on hydraulic fracturing waste management has expanded in recent years, few studies provide field data on reusable containerized systems. Many papers published by SPE acknowledge the drawbacks of open pits and call for mobile, modular, and sustainable alternatives. However, this paper is among the first to quantify:

- Full deployment metrics.
- Waste reduction outcomes.
- ESG implications.
- Reusability cost amortization.

The introduction of this system aligns with broader industry trends seeking closed-loop fluid systems, minimal surface impact, and automated waste handling. The solution aligns closely with the Environmental, Social, and Governance (ESG) principles advocated in modern energy development. As outlined by API standards and SPE technical papers, the integration of reusable, trackable waste handling systems contributes to improved environmental metrics and long-term stakeholder value.

What distinguishes the system discussed in this paper is the field-backed quantification of performance. Unlike many studies that propose theoretical models, this paper presents operational data, including effluent volume tracking, logistical improvements, and site performance benchmarks.

## ESG and HSE Performance

The mobile containment system aligns with global best practices in waste minimization by providing a zero-discharge solution. By completely isolating flowback fluids from soil and air, the system removes the risk of hydrocarbon leaching, which is a common issue with pit-based containment methods. This translates directly to reduced ecological footprint and greater ease in obtaining regulatory approvals. Environmental stewardship in upstream operations is no longer a competitive advantage it is a regulatory and social mandate.

The system delivers measurable contributions to ESG targets:

- **Environmental:** Zero emissions, spills, or landfill waste. The system is designed to minimize its carbon footprint, with no emissions or greenhouse gases released during operation, reducing the overall environmental impact.
- **Social:** Improved safety for workers by eliminating pit hazards. The system eliminates pit hazards, reducing the risk of accidents and injuries to workers, and providing a safer working environment.
- **Governance:** Full documentation of waste handling, traceability, and regulatory adherence. The system provides a complete and accurate record of waste handling, including generation, storage, and disposal, ensuring transparency and accountability. As well as the system allows for the tracking and tracing of waste from generation to disposal, ensuring that all waste is properly handled and disposed of

From an HSE perspective, the system eliminates:

- Personnel exposure to open effluent, as the system closed-loop design eliminates the risk of workers being exposed to open effluent, reducing the risk of illness and injury.
- Fall hazards from pit edges. The system's design eliminates the need for open pits, reducing the risk of falls and other accidents, and creating a safer workplace for employees.
- Burn risks from overflow at high-temperature discharge points. The system's advanced safety features and closed-loop design eliminate the risk of burns and other injuries from high-temperature discharge points.
- Soil contamination and environmental impacts. The system's design and operation minimize the risk of soil contamination and environmental impacts, protecting the environment and reducing the risk of pollution.

## Conclusions

The trial deployment of the subject mobile wastewater containment system successfully met and exceeded all project objectives. Key achievements include:

- 100% effluent containment with no leakage or contamination.
- 2,495 bbls managed over 21 days without operational interruptions per gas well.
- 33% reduction in tanker trips.
- Complete elimination of environmental remediation activities.
- Significant cost savings per well.
- Reusable design contributing to long-term sustainability.
- The system represents a replicable model for modern fracturing operations and offers a viable pathway for workers seeking to modernize their HSE and waste management practices.

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This work complements the broader trend toward minimizing land disturbance and promoting efficient use of resources in completions. It supports the case for industry-wide adoption of closed-loop systems, particularly in regions with fragile ecosystems or heightened regulatory oversight.